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A Computational Method for the Flattening and Fabrication of Compression-Dominant Shell Structures



ABSTRACT

Compression-dominant shell structures achieve high structural efficiency through geometry-driven load transfer, yet their fabrication remains difficult due to the incompatibility between doubly curved geometries and flat sheet-based manufacturing processes. This paper presents a computational tuck-folding method for translating discretized shell geometries into developable fold patterns while preserving reconstructability and fabrication control. Operating on planar-faced polysurfaces generated through computational form-finding workflows, the method introduces localized fold offsets, referred to as tucks, along selected edges during flattening. These tuck-folds compensate for geometric incompatibilities generated during planar unfolding while enabling controllable variation of tuck width for structural stiffening and fabrication constraints. The workflow integrates geometric closure conditions, vertex-based degrees of freedom, and structurally informed tuck-width assignment into a unified computational system implemented in Grasshopper. Tuck widths are constrained by compatibility relationships at vertices, producing a limited but controllable solution space for fold generation. Compression-force data extracted from the shell form-finding process informs tuck-width distribution within fabrication constraints such as material thickness and machine bed dimensions. Two case studies are presented: physical paper models validating geometric reconstruction, foldability, and stiffness, and a concrete shell formwork proposal using CNC-cut plywood and fabric membranes. The research contributes a computational folding method that bridges geometric form-finding, structurally informed discretization, and materially efficient fabrication workflows for freeform shell construction.

1 Proposed Design for a Modular Outdoor Gallery Concrete Pavilion

INTRODUCTION

Compression-dominant shell structures gain material efficiency when geometry aligns load paths with axial compression rather than bending, as Block and Ochsendorf [2007] demonstrated through thrust-network analysis. Nejur and Akbarzadeh [2021] extended this equilibrium logic into computational 3D graphic statics (3DGS), where force and form diagrams remain coupled during geometric exploration. These methods make compression-dominant shells powerful design objects because structural behavior is generated through geometry rather than corrected after form-making.

Construction weakens this link when doubly curved shell geometries must be translated into flat stock, transportable parts, and controllable assembly procedures. As Hawkins et al. [2016] reported, flexible formwork research emerged partly because conventional rigid molds often turn geometric complexity into bespoke tooling, intensive labor, and high material consumption. A subtractive workflow such as CNC-milled plywood formwork can provide geometric fidelity, but in this paper it operates as a material-intensive counterexample rather than the proposed fabrication strategy.

Veenendaal and Block [2014], Méndez Echenagucia et al. [2019], and Papescu et al. [2021] also demonstrated that cable-net, fabric, and knitted formwork systems can reduce the mass of shell formwork by transferring geometric control into lightweight tensioned or textile assemblies. These systems show that shell fabrication does not need to depend on heavy, rigid molds, yet their accuracy still relies on specialized tensioning setups, textile fabrication, or construction-specific support infrastructures. Paper-based sheet formwork offers a different route: it uses the flatness, foldability, and low mass of sheet material as the primary geometric carrier rather than as a secondary template attached to a separate mold.

The central difficulty is that flattening a discretized shell is not a neutral drafting operation. A planar face can be unfolded individually, but a closed neighborhood around a spatial vertex cannot always preserve all three-dimensional adjacencies in the plane, which creates gaps, overlaps, or rotational mismatch during reconstruction. Conventional tabs and flanges reconnect cut parts after flattening, but they treat assembly as a fastening problem rather than encoding curvature, stiffness, or structural information into the pattern itself. Tachi [2010] indicated that in computational origami, hidden tucks can absorb geometric incompatibilities between a target polyhedral surface and a planar crease pattern. As Dias et al. [2012] showed, crease geometry also affects mechanical behavior, which allows folds to operate as stiffness-producing features rather than neutral hinge lines.

This paper develops tuck-folding as a computational method for the flattening and fabrication of compression-dominant shell structures. The method introduces localized fold offsets, referred to as tucks, along selected edges of a planar-faced shell discretization. These tucks absorb flattening incompatibilities while preserving reconstructability, and their widths become controllable parameters linked to vertex closure, material

thickness, sheet size, and force distribution. Diarte Almada and Shaffer [2025] reported in cardboard formwork experiments that paper-based molds can participate in concrete fabrication while exposing critical constraints such as waterproofing, dimensional control, and casting load. This research, therefore, treats paper sheet material not as a representational model, but as a temporary geometric carrier for thin-shell formwork.

The workflow begins with a compression-dominant shell generated through graphic statics-based form-finding. The shell is converted into a planar-faced polysurface, segmented into fabrication-aware regions, and flattened into two-dimensional patterns. Tucks are inserted where unfolding creates geometric incompatibility, and their widths are constrained by closure relationships at vertices. Compression-force data from the form-finding process informs the distribution of tuck widths within fabrication limits such as material thickness, sheet size, and assembly sequence. The resulting paper sheets are cut, creased, folded, and assembled as lightweight formwork for casting a thin concrete shell.

The paper follows the same sequence as the method: shell generation defines the compression-dominant target geometry; segmentation and flattening define the fabrication domain; tuck-folding formalizes vertex closure and tuck-width control; prototype studies evaluate paper reconstruction, stiffness, and paper-sheet formwork behavior; the discussion identifies tolerance, waterproofing, casting pressure, demolding, and scale-up constraints that determine the method's architectural applicability.

LITERATURE REVIEW

Related studies relevant to tuck-folding for compression-dominant shell structures begins with equilibrium-based shell design. Block and Ochsendorf [2007] reformulated historical funicular form-finding as a graphical method for generating compression-only vaulted surfaces inside prescribed geometric envelopes. Akbarzadeh et al. [2015] extended this line of research to polyhedral systems by establishing the reciprocal relationship between funicular polyhedral frames and convex force diagrams, making discrete planar-faced geometries structurally legible rather than treating them as post-rationalized meshes. Nejur and Akbarzadeh [2021] translated these ideas into a computational workflow that preserves the coupling between form and force during design exploration. Together, these studies explain how compression-dominant shell geometry can be generated with structural logic embedded from the outset, but they leave open the problem of how such geometries should be materialized through flat, lightweight sheet fabrication.

A second body of research addresses construction through lightweight formwork. In their review of flexible formwork technologies, Hawkins et al. [2016] showed that non-rigid molds can reduce material demand while widening the geometric range of concrete construction. Within shell fabrication more specifically, Veenendaal and Block [2014] developed a hybrid cable-net and fabric workflow that integrated form-finding, structural analysis, and formwork design for thin concrete shells. Méndez Echenagucia et al. [2019] demonstrated the feasibility of this approach

at full scale through a cable-net and fabric formwork prototype that achieved doubly curved shell geometry with a largely reusable support system. Popescu et al. [2021] expanded the same trajectory through KnitCandela, where cable-net and knitted stay-in-place formwork enabled a materially light concrete shell assembled from digitally fabricated textile components. This literature shows that shell construction does not require heavy rigid molds, but geometric control in these systems still depends on calibrated tension networks, custom textile production, or project-specific scaffolding infrastructures.

A third strand of work concerns fabrication-aware rationalization. Eigensatz et al. [2010] framed the paneling of architectural freeform surfaces as an optimization problem balancing geometric deviation, mold reuse, and production cost. That contribution is important because it shifted discretization from a purely geometric task to a manufacturing problem. For the present research, however, panelization alone is insufficient. It can rationalize a shell into planar or nearly planar parts, yet it does not explain how flattening incompatibilities at vertices and along shared edges should be resolved when a shell must be reconstructed from foldable sheet material. The literature on architectural paneling therefore provides a precedent for fabrication-aware subdivision, but not for the fold-based compensation strategies required by sheet-formwork systems.

Computational origami addresses that missing geometric problem more directly. Tachi [2010] showed that a target polyhedral surface can be embedded into a planar crease pattern through hidden tucks that absorb incompatibilities generated during flattening while preserving one-sheet reconstructability. Dias et al. [2012] added a mechanical dimension by showing that crease geometry affects stiffness and deformation, which means folds can operate as structural features rather than as neutral joints. These studies are highly relevant because they shift the role of folding from connection to geometric compensation and mechanical control. Their main limitation in relation to shell construction is disciplinary: they primarily target geometric reconstruction, deployability, or transformability rather than temporary formwork for concrete casting.

Recent work on paper-derived molds begins to close that material gap. Diarte Almada and Shaffer [2025] demonstrated that corrugated cardboard can serve as formwork for precast concrete components and identified the practical constraints that govern such systems, including waterproofing, dimensional stability, and resistance to casting pressure. This contribution establishes paper-based material as a credible geometric carrier for concrete fabrication rather than only a representational medium. What remains missing across the literature is an integrated workflow that connects force-informed shell generation, fabrication-aware discretization, tuck-based flattening, and paper-sheet formwork. Existing studies explain how to find compression-dominant forms, how to lighten shell formwork, how to rationalize freeform surfaces, and how to compensate flattening through folds. They do not yet provide a computational method that combines these operations for the fabrication of compression-dominant shell structures from foldable sheet material.

METHODS

To demonstrate the tuck-folding method, this section develops a complete workflow for a modular floor slab system — a compression-dominant funicular unit composed of a column, a canopy, and the arches connecting them. The workflow proceeds in three stages: form-finding with 3DGS generates the target geometry as a planar-faced polysurface with known edge forces (Form Finding); the tuck-folding algorithm translates that geometry into a flat folding pattern with controllable tuck widths (Flattened Pattern Algorithm); and material-specific processing converts the pattern into fabrication files for thin or thick sheet stock (Fabrication) as shown in Figure 2.

Form Finding

The target structures are generated using PolyFrame 2, a geometry-based structural form-finding plugin for Rhinoceros 3D and Grasshopper that implements polyhedron-based 3D graphic statics Lu et al. [2024]. Through this tool, the base geometry and subdivision pattern of a force diagram are manipulated to generate its reciprocal form diagram, which defines the target structure. For a modular floor slab system, the external forces are identified as vertical loads, lateral supports from neighboring modules, and vertical support from the bottom column, as shown in Figure 2(a)[1]. Given the directional nature of these forces, the base geometry of the force diagram is determined as shown in Figure 2(a)[2].

Subdividing the three force cells that compose the force diagram produces more complex structures in the reciprocal form diagram. In this design, a distinct subdivision method is applied to the column, canopy, and arch force cells. While the column and canopy force cells each consist of a single external force surface, the arch force cell consists of all internal force surfaces that connect the former two.

The column cell takes all vertical support from the bottom. Subdividing the bottom surface of the cell controls the plan profile of the column, while vertical subdivisions control the curvature of its section profile.

The canopy cells take all vertical load from above. Subdividing the top surface produces different canopy patterns; Figure 3 shows four such patterns as basic subdivision typologies Zheng et al. [2020]. Extruding each surface segment toward different vertices on the column cell also produces different arch positions.

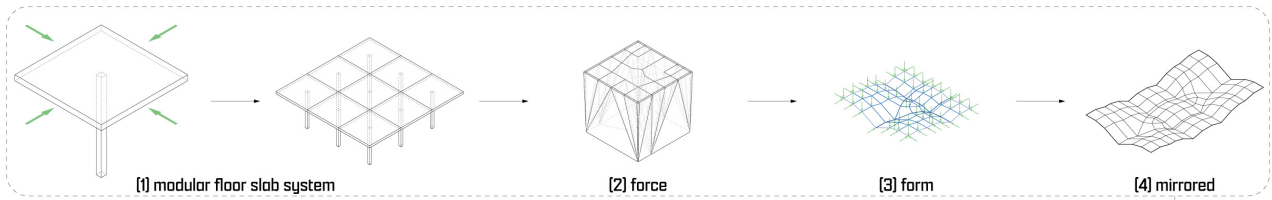
Subdivisions of the arch cell should follow the subdivision patterns of the column and canopy cells. Because the arch cell always takes the following shape, its resulting curvature in the form diagram is always anticlastic.

To preserve a compression-dominant structure without tension and to maintain the column support location of the slab, point, line, or plane constraints are applied to the form diagram to achieve a closer approximation of the given force diagram.

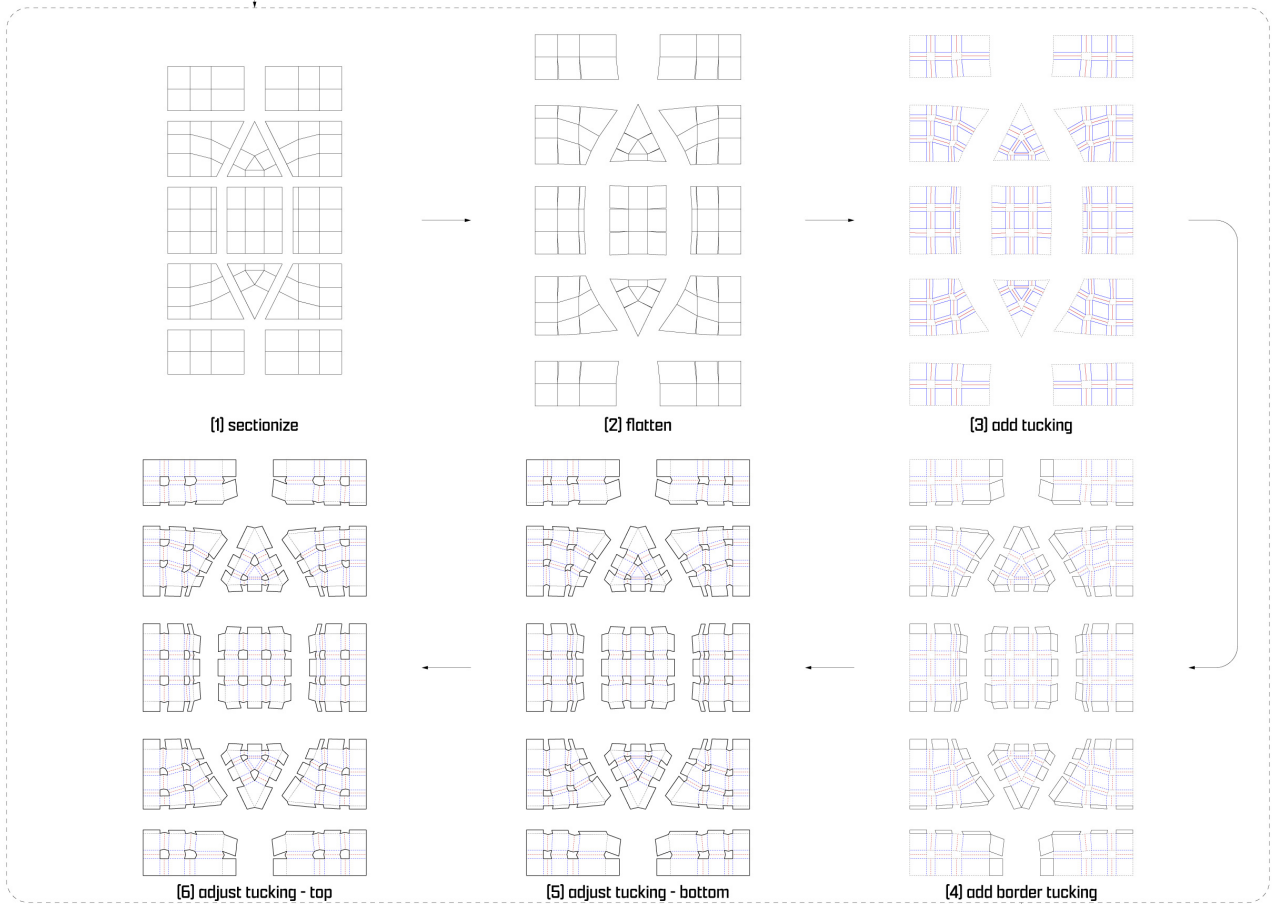
Flattened Pattern Algorithm

The flattening algorithm converts the target shell into a two-dimensional

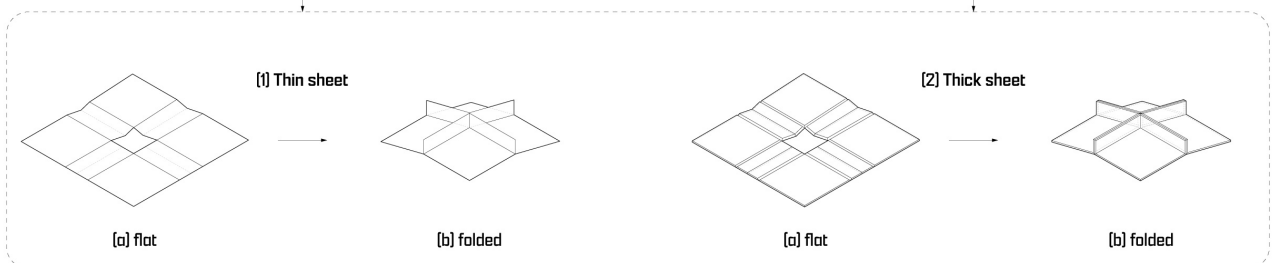
(a) Form-Finding

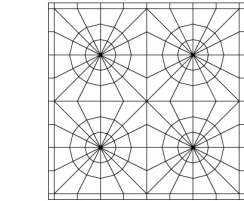


(b) Tuck-Folding

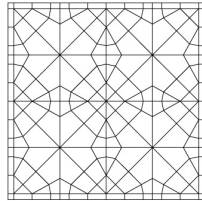


(c) Fabrication

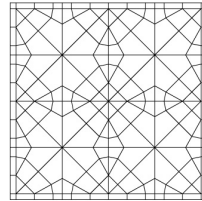




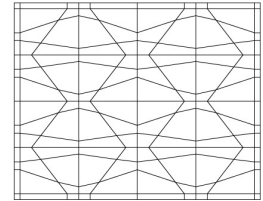
(a1)



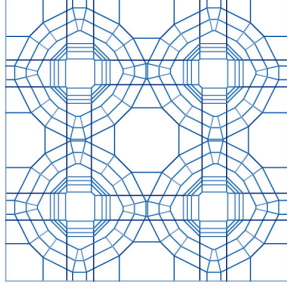
(b1)



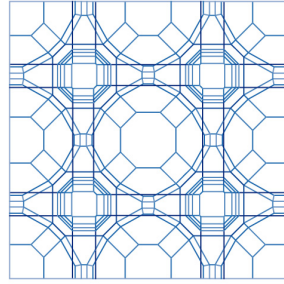
(c1)



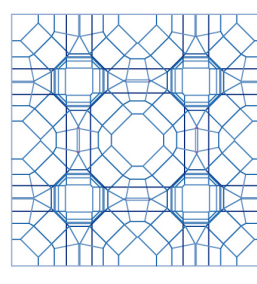
(d1)



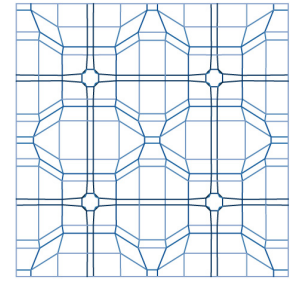
(a2)



(b2)



(c2)



(d2)

3 Subdivision Typology Study

fold pattern that can be cut and refolded to reconstruct the original geometry. Because the fold strategy depends on the curvature at each vertex, the two curvature typologies identified during form-finding are considered first.

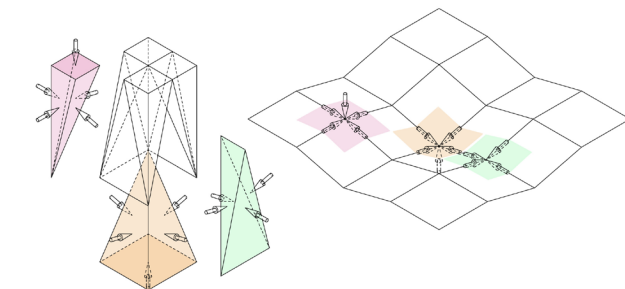
Different force typologies result in different form curvatures. While the column and canopy force cells always produce synclastic curvature, the arch force cells always produce anticlastic curvature, as shown in Figure 4. At individual vertices in the form diagram, synclastic vertices have an inner corner sum of less than 360 degrees, which makes them easier to flatten than anticlastic vertices. In addition, the edges of the arch form diagram always carry higher forces.

Given the force and curvature analysis above, the slab geometry is sliced primarily along the arch, as shown in the example in Figure 2(b)(1), in order to (1) simplify the subsequent flattening steps, (2) give full control over the tuck thickness corresponding to the force carried by each edge, and

(3) accommodate the fabrication-area limitations of laser cutters, CNC machines, and 3D printer beds. Secondary splits are made when individual pieces remain larger than the fabrication-area constraint.

Flattening a curved polysurface into a two-dimensional pattern introduces geometric incompatibilities, because the shared-edge relationships that hold in three-dimensional space cannot be preserved directly in the plane. At least one edge discontinuity must therefore be introduced during the unfolding process. The discontinuous edges can be rejoined by adding an isosceles triangular tuck whose angular bisector is folded by 180 degrees, reconstructing the curved polysurface as shown in Figure 5.

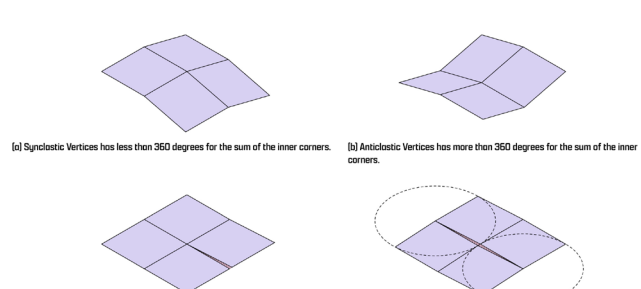
To enable more precise control over the folded geometry, inner tucks are added to all connected edges in addition to the existing isosceles triangular tuck, which is the minimum tuck required to produce a flat folding pattern.



(a) Basic force diagram of a floor slab module with one column cell in orange, four arch cells in green, and four canopy cells in pink.

(b) Corresponding form diagram with the vertices relative to the cells on the left. We can see that the canopy column cells results in synclastic curvature, where as the arch cells results in anticlastic curvature

4 Force Typology and Form Curvature Relationship



(a) Synclastic Vertices has less than 360 degrees for the sum of the inner corners.

(b) Anticlastic Vertices has more than 360 degrees for the sum of the inner corners.

(a) Basic force diagram of a floor slab module with one column cell in orange, four arch cells in green, and four canopy cells in pink.

(b) Corresponding form diagram with the vertices relative to the cells on the left. We can see that the canopy column cells results in synclastic curvature, where as the arch cells results in anticlastic curvature

5 Flattening Logic of a Synclastic Vertex and an Anticlastic Vertex

For a vertex with three joining surfaces, the degree of freedom for adding inner tucks is one, meaning that once the tuck thickness of one edge is determined, the other two are thickened correspondingly. For a vertex with four joining surfaces, the degree of freedom is two, meaning that the thickness of two selected edges can be controlled while the other two are thickened correspondingly. From this principle, the degree of freedom at each vertex, given its number of joining edges M , is $M - 2$.

Neighboring vertices share an edge. For two neighboring vertices, where vertex A has M joining edges and vertex B has N joining edges, the total degree of freedom is $M + N - 5$. Based on the degree-of-freedom principle for tuck addition proposed above, the edges to be constrained can be selected manually to generate the inner-tuck thickness pattern shown in Figure 2(b)(3). Border tucks are independent of one another, since adding them creates no geometric conflicts. They can therefore be made as thick as the material requires to provide the necessary forces at the splitting edges, as shown in Figure 2(b)(4).

The shorter edges of the tuck areas must be adjusted to specific angles to preserve geometric stiffness and fidelity after folding. Figure 2(b)(5) shows the adjusted tuck geometry when the tucks are folded upward, and Figure 2(b)(6) shows the adjusted geometry when the tucks are folded downward.

Fabrication

The choice of formwork material determines how the fabrication files are generated. The two material categories differ in one geometric respect: whether the sheet thickness is small enough to be ignored at the fold lines. This distinction governs both the fabrication method and the way the fold edges are encoded.

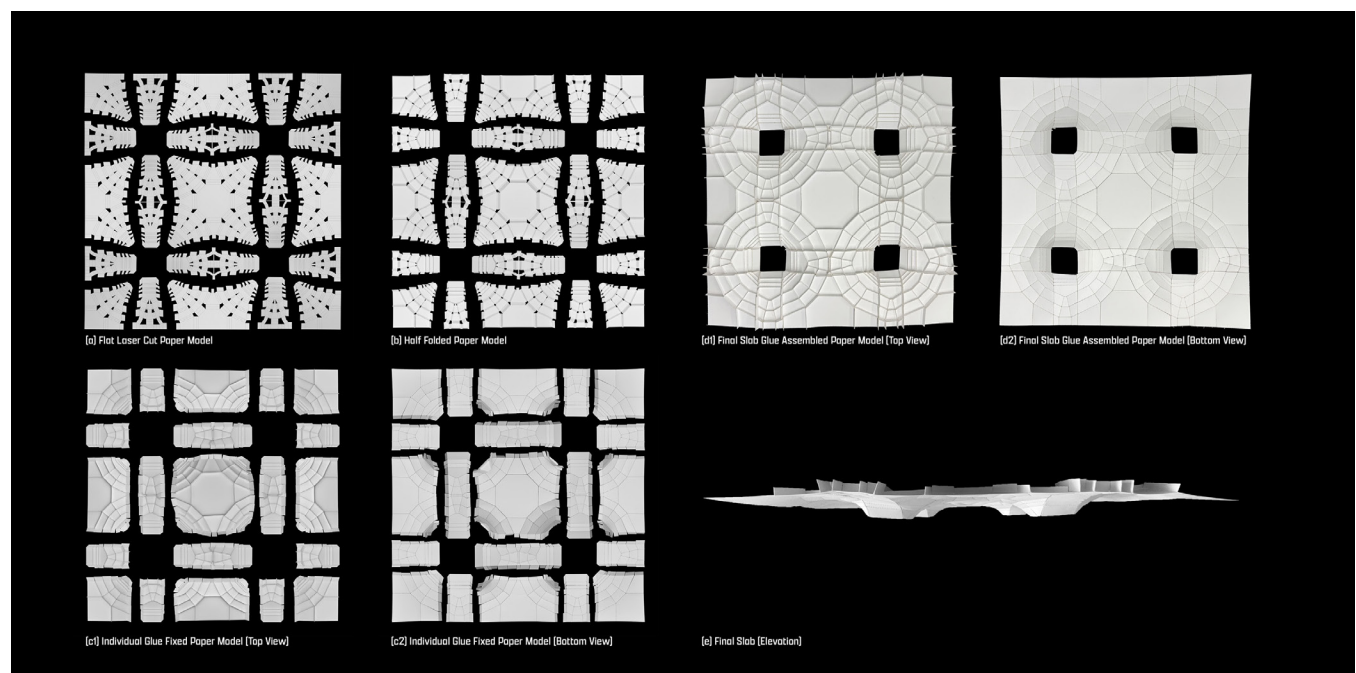
Thin sheet materials, such as paper or sheet metal Akbari et al. (2021), are thin enough that their thickness can be ignored during folding. The two faces meeting at a fold rotate about a single shared line without colliding, so a fold can be represented as a zero-width crease. The fabrication files are produced by laser cutting, with the folded edges scored as dashed lines that act as precise fold lines for on-site assembly shown in Figure 2(c)(1).

Thick sheet materials, such as 3D prints on fabric or plywood panels riveted to fabric, cannot be treated this way. Because the material has appreciable depth, the two faces collide along the inner edge as they rotate, and a zero-width crease is no longer foldable. The fold must instead be opened into a physical crease: each edge is beveled at an angle corresponding to its fold angle, so that the two faces close cleanly, as shown in the example in Figure 2(c)(2). The beveled crease geometry is produced by 3D printing or CNC milling, while a continuous fabric layer bonded across the joints provides a thin, reusable hinge at each edge. The bisector fold edge of the tuck always folds outward and requires no beveling, whereas the shorter edges of the tuck are beveled to match the converging geometric constraints at the vertices.

RESULTS AND DISCUSSION

The tuck-folding method is evaluated through two case studies, one for each of the fabrication approaches introduced in the previous section. The first uses thin sheet material to verify that the flattening algorithm produces a pattern that folds back into the target slab geometry. The second applies the thick-sheet approach to a concrete pavilion proposal, testing the method at architectural scale.

Case Study 1: Paper Model — Proof of Geometry



6 Case Study 1 - 1:10 Paper Folding Model

In the first case study, a paper prototype is used to verify that the tuck-folding pattern generated by the flattening algorithm folds back into the target floor slab geometry. Two aspects of the paper model are examined to assess the success of the computational tool:

- the fidelity of the folded geometry relative to the target geometry;
- the stability of the assembled model.

The first step in fabricating the paper prototype is to generate laser-cutting files from the folding pattern produced by the computational tool. The outlines of the pieces are converted into continuous lines, while the mountain and valley creases are converted into dashed lines — a series of short segments — to ease folding, as shown in Figure 6(a). All lines are executed as cuts by the laser cutter.

Scoring the fold creases proved less effective than dashed line cuts in this case. The pattern contains both mountain and valley creases, which fold in opposite directions; a scored crease folds easily only away from its scored face, so accommodating both crease types would require scoring on both sides of the paper. Aligning the two scoring passes reduces accuracy, so dashed line cuts, which fold equally well in either direction, were preferred.

The next step is to fold the mountain and valley creases in their correct directions, as shown in Figure 6(b).

The third step is to secure the geometry by gluing the tucks together. Each tuck consists of one mountain crease and two valley creases. The mountain crease always folds by 180° , while the valley creases fold at varying angles. Gluing the two inner surfaces of each tuck fixes the 180° fold of the

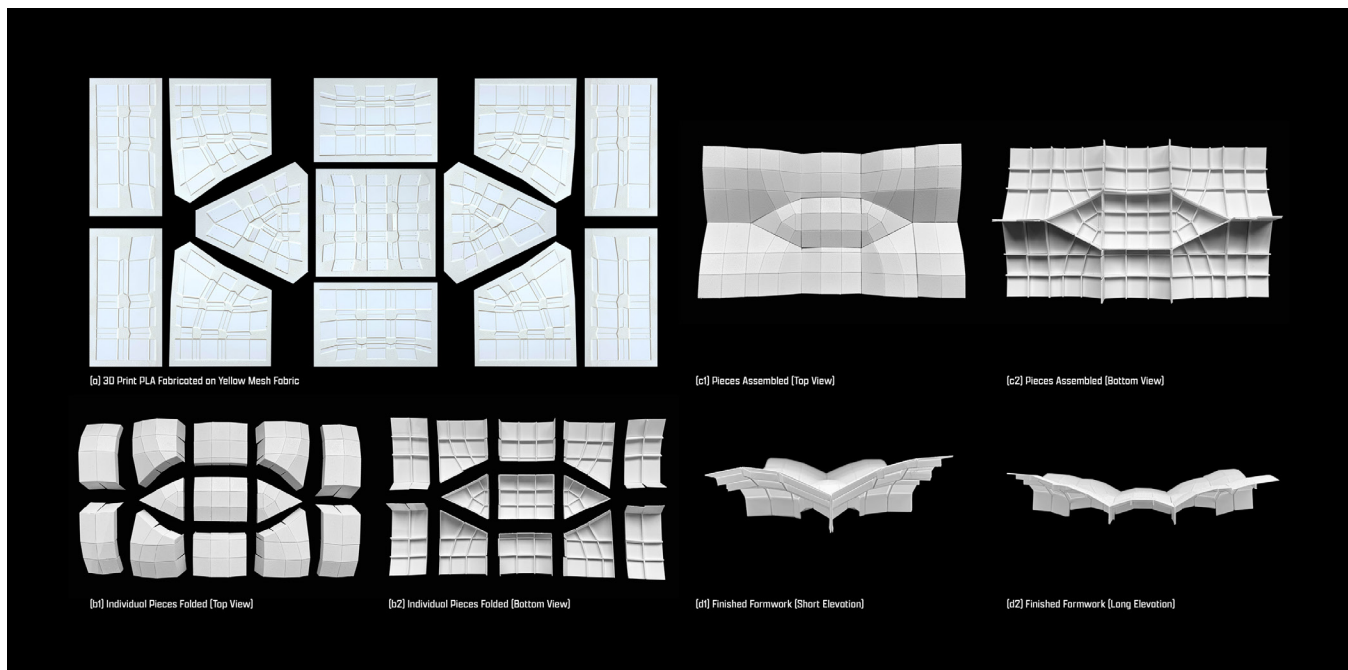
mountain crease, as shown in Figure 6(c1) and (c2).

The final step is to glue the border tucks together, assembling the complete slab geometry from the individual tuck-folded pieces, as shown in Figure 6(d1) and (d2).

The completed model was evaluated against the two criteria defined above: geometric fidelity and stability. Fidelity was assessed by overlaying the physical model on the digital target geometry. The folded prototype reproduced the target slab geometry with high resemblance, confirming that the flattening algorithm generates a pattern that folds back into the intended shape. The model was fabricated from Bristol drawing paper and measured 650 mm x 650 mm at 1:10 scale, assembled from 21 pieces, each carrying a continuous set of tucks. Assembly was straightforward.

Fidelity was weakest in two conditions: where the tucks were too thin, and where a tuck was much longer than it was wide. In both cases the tuck was less able to hold its intended fold, locally reducing the accuracy of the reconstructed surface. This is consistent with the tuck-width logic developed in the previous section, where tuck thickness is tied to the force carried at each edge — under-scaled tucks correspond to the regions where geometric control is least constrained.

Stability, in the sense of stiffness, depended on fixing the tucks. Once all tuck edges were glued, the mountain creases held their 180° fold and the model was rigid and dimensionally stable, with no observable spring-back. The weakest region was the model's outer edge: as a single module, the prototype relies on neighboring modules to provide horizontal support at its boundary, and required external horizontal support to remain stable in isolation. Gluing also introduces a practical limitation — it fixes the



7 Case Study 2 - 1:25 Folded Thick-Sheet Formwork for Concrete Casting

geometry but is not reusable, so while it is sufficient to prove the folded geometry, it is not a viable fixing method for reusable formwork. This motivates the fabric-hinge approach explored in Case Study 2.

Overall, the paper prototype demonstrates that the tuck-folding pattern generated by the flattening algorithm folds successfully into the target slab geometry, validating the geometric basis of the method.

Case Study 2: Folded Thick-Sheet Formwork for Concrete Casting

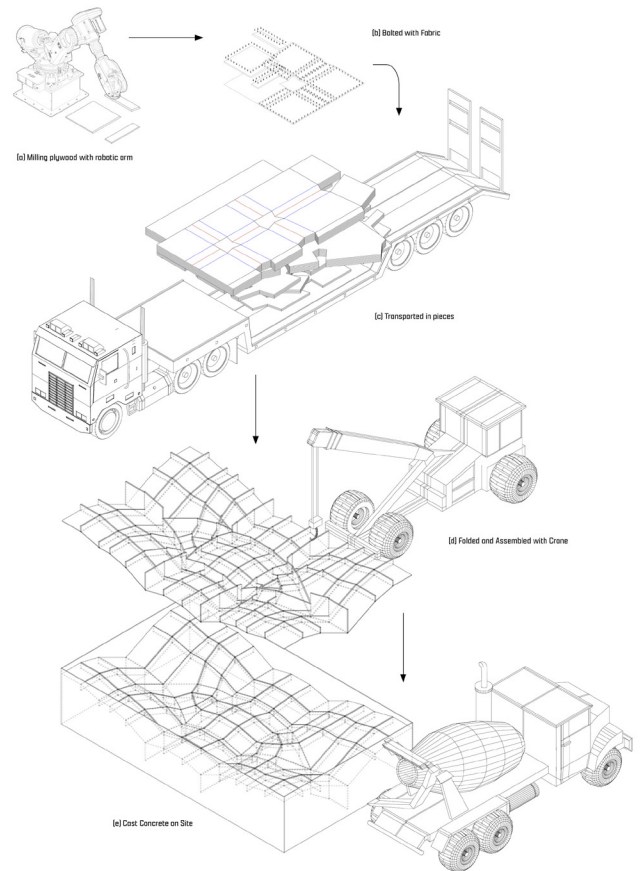
The second case study applies tuck-folding as a thick-sheet formwork system for concrete shell fabrication, developed through a design proposal for a concrete pavilion. The pavilion is a modular funicular shell spanning 10 m by 5 m, which can be repeated into a floor slab system or stand alone as a single canopy with lateral bracing, as shown in Figure 9.

For fabrication, the shell is divided into 13 pieces along the anticlastic vertices, following the splitting logic used in Case Study 1 and respecting the print-bed size limits of the fabrication equipment. Dividing along the anticlastic vertices isolates the regions of highest curvature and force, where the previous case study showed geometric control to be most demanding.

To validate the thick-sheet approach, a formwork prototype was fabricated at 1:25 scale as shown in Figure 7. Each panel consists of a continuous layer of gold tulle mesh fabric with PLA components 3D-printed directly onto it, carrying the angled crease geometry on the upper face. The fabric forms a single continuous layer across all creases, providing a thin, reusable hinge at each fold edge, while the printed panels supply the beveled crease geometry required for thick-sheet folding. The 180° fold creases were printed with a 0.1 mm offset on each side to prevent the two faces from fusing during printing, since they meet flush in the flat pattern.

As a formwork-fabrication demonstration, the prototype was not used to cast concrete; instead, it verifies the geometry of thick-sheet folding along the same two criteria as Case Study 1 — fidelity to the target geometry and stability once assembled. The printed panels folded cleanly to the target curvature and were simple to operate, and the combination of the fabric layer and the printed crease angles held the intended fold angles at each edge. Because the fabric hinge allows the panels to fold and unfold without damage, the assembled formwork is reusable — addressing the fixing limitation identified in Case Study 1, where the glued paper model could not be reused.

At full scale, we propose a workflow that combines CNC-cut plywood ribs with fabric membranes to produce reusable folded formwork components capable of shaping high-fidelity funicular shell geometries, as shown in Figure 8. By embedding geometric compensation and structural reinforcement directly into the fold pattern, the method reduces the number of highly customized formwork elements typically required for freeform concrete casting. The folded components can also be compactly transported and reassembled, improving fabrication efficiency and reducing material waste. Full-scale fabrication in plywood, and casting concrete



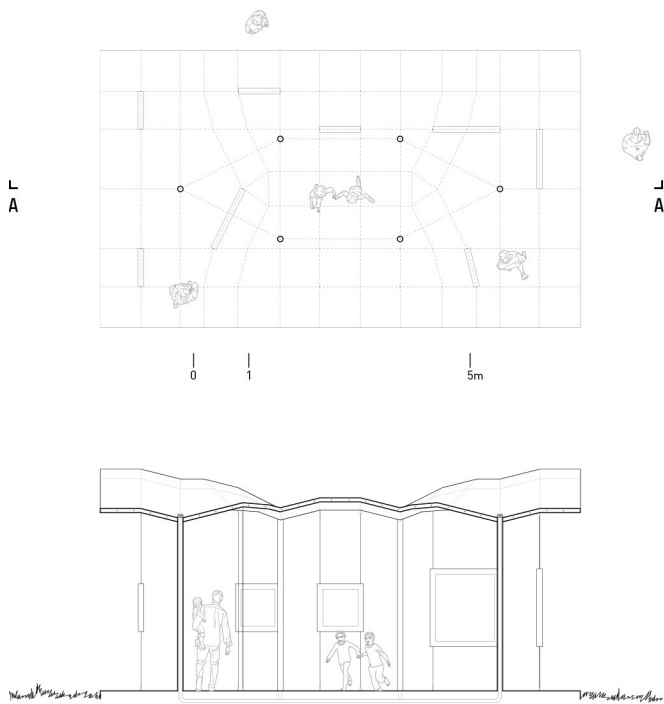
8 Proposed Fabrication and Construction Workflow

against the assembled formwork, are identified as directions for future work.

Together, the two case studies demonstrate that tuck-folding operates as both a geometric compensation strategy and a fabrication-aware discretization workflow: the paper model verifies that the flattened pattern reconstructs the target geometry, and the pavilion prototype shows that the same principle extends to reusable thick-sheet formwork at architectural scale. Encoding stiffness into the fold pattern remains a design goal of the method rather than a validated result, and is identified as a direction for future work. While the current implementation focuses on funicular shell geometries, the method can be extended to other freeform discretized surfaces.

CONCLUSION

This research addressed a central fabrication problem in compression-dominant shell construction: structurally efficient doubly curved geometries are difficult to translate into flat, lightweight, and controllable fabrication systems without losing geometric continuity or relying on material-intensive custom molds. In response, the paper developed a computational tuck-folding method that inserts localized fold offsets along selected edges of planar-faced shell discretizations. Rather than treating unfolding discontinuities as secondary fastening details, the method uses tuck-folds to absorb flattening incompatibilities, preserve



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reconstructability, and embed fabrication and structural logic directly into the two-dimensional pattern.

The study showed that tuck-folding can operate across several connected levels. Geometrically, tuck insertion compensates for gaps, overlaps, and rotational mismatch created during the flattening of discretized shell geometries. Computationally, tuck-width assignment is organized through vertex-based closure constraints, where a limited number of independent variables define a controllable design space. Structurally, compression-force information extracted from the form-finding process can guide tuck-width distribution, allowing the fold pattern to respond to local force intensity while remaining within fabrication limits. Material constraints such as sheet thickness, foldability, collision avoidance, machine-bed dimensions, and assembly sequence further shape the feasible tuck pattern.

The two case studies indicate the potential of the method as a bridge between funicular form-finding and material fabrication. The paper model study validated the basic foldability and reconstructability of tuck-folded patterns, while also showing that inserted tucks can increase edge stiffness and reduce local deformation. The folded formwork study extends the workflow toward thin-shell concrete fabrication by positioning sheet-based folded components as temporary geometric carriers rather than merely representational models. In this sense, the contribution of this research is not only a flattening technique, but a fabrication-aware framework that connects shell geometry, tuck-based unfolding, stiffness

control, and lightweight formwork assembly.

The current work also has limitations. The geometric constraint system requires further formalization, particularly in relation to explicit closure equations, the selection of independent tuck-width variables, and the projection of force-based width targets into the feasible solution space. The physical validation remains preliminary and requires more quantitative evaluation of geometric deviation, stiffness improvement, tolerance accumulation, and fabrication accuracy. For concrete formwork applications, additional challenges such as waterproofing, casting pressure, demolding, material durability, and scale-up must be tested before the method can be considered construction-ready.

Future work will therefore focus on three directions. First, the computational model should be expanded into a more rigorous solver for tuck-width optimization under geometric, structural, and fabrication constraints. Second, the workflow should be integrated with structural simulation to evaluate the relationship between tuck geometry, shell stiffness, and casting performance. Third, larger-scale physical prototypes should be developed to test robotic or assisted folding processes, reusable modular formwork systems, and applications beyond funicular shell geometries.

By transforming flattening errors into controllable fold features, tuck-folding reframes sheet-based fabrication as an active design instrument for compression-dominant shell construction. The method suggests that the gap between computational shell form-finding and material realization

can be narrowed not by simplifying geometry after design, but by embedding geometric compensation, structural response, and fabrication constraints into the fold pattern itself.

REFERENCES

Akbari, M., Y. Lu, and M. Akbarzadeh [2021]. From design to the fabrication of shellular funicular structures. In *Proceedings of the Association for Computer-Aided Design in Architecture [ACADIA]*.

Akbarzadeh, M., T. Van Mele, and P. Black [2015]. On the equilibrium of funicular polyhedral frames and convex polyhedral force diagrams. *Computer-Aided Design* 63, 118-128.

Black, P. and J. Ochsendorf [2007]. Thrust network analysis: A new methodology for three-dimensional equilibrium. *Journal of the International Association for Shell and Spatial Structures* 48(3), 167-173.

Diarte Almada, J. C. and M. Shaffer [2025]. Experimenting with cardboard formwork for precast concrete building components. *Journal of Architectural Engineering* 31(3), 04025022.

Dias, M. A., L. H. Dudte, L. Mahadevan, and C. D. Santangelo [2012]. Geometric mechanics of curved crease origami. *Physical Review Letters* 109(11), 114301.

Eigensatz, M., M. Kilian, A. Schiftner, N. J. Mitra, H. Pottmann, and M. Pauly [2010]. Paneling architectural freeform surfaces. *ACM Transactions on Graphics* 29(4), 1-10.

Hawkins, W. J., M. Herrmann, T. J. Ibell, B. Kromoser, A. Michalski, J. J. Orr, P. Shepherd, D. Veenendaal, and M. West [2016]. Flexible formwork technologies: A state of the art review. *Structural Concrete* 17(6), 911-935.

Lu, Y., M. Hobblicsek, and M. Akbarzadeh [2024]. Algebraic 3d graphic statics with edge and vertex constraints: A comprehensive approach to extending the solution space for polyhedral form-finding. *Computer-Aided Design* 166, 103620.

Méndez Echenagucia, T., D. Pigram, A. Liew, T. Van Mele, and P. Black [2019]. A cable-net and fabric formwork system for the construction of concrete shells: Design, fabrication and construction of a full scale prototype. *Structures* 18, 72-82.

Nejur, A. and M. Akbarzadeh [2021]. Polyframe, efficient computation for 3d graphic statics. *Computer-Aided Design* 134, 103003.

Papescu, M., M. Rippmann, A. Liew, L. Reiter, R. J. Flatt, T. Van Mele, and P. Black [2021]. Structural design, digital fabrication and construction of the cable-net and knitted formwork of the knitcandela concrete shell. *Structures* 31, 1287-1299.

Tachi, T. [2010]. Origamizing polyhedral surfaces. *IEEE Transactions on Visualization and Computer Graphics* 16(2), 298-311.

Veenendaal, D. and P. Black [2014]. Design process for prototype concrete shells using a hybrid cable-net and fabric formwork. *Engineering Structures* 75, 39-50.

Zheng, H., X. Wang, Z. Qi, S. Sun, and M. Akbarzadeh [2020]. Generating and optimizing a funicular arch floor structure. In *Proceedings of the 40th Annual Conference of the Association for Computer-Aided Design in Architecture [ACADIA]*.

[Conference Paper] Cremers, Jan. 2011. "Energy Saving Design of Membrane Building Envelope." In *International Conference on Textile Composites and Inflatable Structures: Structural Membranes 2011*, edited by E. Oñate, B. Kröplin and K.-U. Bletzinger, 148-155. Barcelona: CIMNE.

IMAGE CREDITS

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